Report on Prompt Experimentation with VeganBot

Objective

The task was to test a structured prompt for a VeganBot ordering assistant. We experimented with different user prompts, analyzed GPT's responses, and summarized the findings in terms of strengths, weaknesses, and lessons learned.

Prompt Variations

Version 1: Simple Order

User asked: 'I want a large vegan burger with avocado.'
GPT correctly identified the menu item, size, and extra. It added the right price and confirmed the order.
✓ Worked well.

Version 2: Ambiguous Request

User asked: 'Give me the usual with a drink.'
GPT hallucinated a 'default order' concept that was not in the schema. It tried to invent a combination meal.
■ Did not work well due to hallucination.

Version 3: Multi-Item Complex Order

User asked: 'Add a medium vegan pizza with mushrooms and a sparkling water.'
GPT correctly parsed both the pizza (with extra) and the drink. The order was formatted with correct pricing.
✓ Worked correctly.

Version 4: Edge Case

User asked: 'Do you have chicken nuggets?'
GPT wrongly attempted to add 'vegan nuggets' which do not exist in the menu, instead of replying that the item is unavailable.
■ Incorrect — hallucinated an item.

Findings

- Strengths: GPT reliably handled menu items, sizes, and extras when explicitly mentioned.
- Weaknesses: GPT sometimes hallucinated (e.g., invented default orders or items not in the menu).
- Creative prompts exposed limits — ambiguous instructions or off-menu requests often failed.

Lessons Learned

1. Clear prompts yield better results — GPT works best when the order matches the menu structure.
2. Ambiguity leads to hallucination — GPT may invent menu items or behaviors that don’t exist.
3. System instructions are crucial — explicitly telling GPT to reject unknown items would improve reliability.
4. Creativity helps identify weaknesses — testing unusual scenarios highlighted where GPT was less reliable.

Conclusion

Experimenting with VeganBot prompts showed that GPT can accurately process structured food orders but struggles with ambiguity and off-menu requests. A clear schema and precise instructions are key to building reliable ordering assistants.